

Name (Student 1): Devansh Chaturvedi

Roll No: 240340

Name (Student 2): Sthit Pragye

Roll No: 241045

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Question: 2

a)

(i) $f_s = 250 \text{ Hz} = 250 \text{ samples/sec}$

$N = 5000 \text{ samples}$

Time duration = $N/f_s = 5000/250 \text{ sec} = 20 \text{ sec}$

(ii) $T_0 = 0.8 \text{ sec}$

$f_0 = 1/0.8 \text{ beats/sec} = 60/0.8 \text{ beats/min} = 75 \text{ beats/min}$

Number of samples = (Time Duration)*(Number of samples in 1 sec) = $0.8 * 250$
= 200 samples

(iii) $f_0 = 1/0.8 \text{ Hz} = 1.25 \text{ Hz}$

b)

(i) Since the signal is periodic in time domain, it appears as an impulse train in the frequency domain with impulses at integral multiples of fundamental frequencies (... $-2f_0$, $-f_0$, 0, f_0 , $2f_0$, ...)

(ii) The QRS spike is responsible for the higher frequency content of ECG. This is because to represent the spikes, a higher frequency sinusoid would be required which can oscillate fast enough to represent the changes in the signal in time domain.

(iii) If the heart rate rises to 150 bpm, then

$f_0 = 150/60 \text{ beats/sec} = 2.5 \text{ Hz}$ i.e. f_0 doubles

The spacing between the components in frequency spectrum (equal to f_0) also doubles from 1.25 Hz to 2.5 Hz.

c)

- (i) To capture one full heartbeat, the width of the window should be equal to 200 samples i.e. duration of the heartbeat. It should be placed at the start of a healthy heartbeat (within the first few seconds) to capture a P wave, QRS complex and the trailing T wave.
- (ii) In the first case, the window width (80 samples) is less than duration of one full heartbeat (200 samples). So, it only captures some fraction of a single heartbeat and truncates the rest. If the window width is 600 samples i.e. more than the width of heartbeat signal, it will capture more than one heartbeat (approx. 3 if placed at beginning of the heartbeat)
- (iii) If the window width is too less, then in time domain, it might truncate off some part of heartbeat i.e. a P wave or a T wave. In the frequency domain, this causes severe spectral leakage. The sharp harmonic impulses widen dramatically, blur together, and spill into neighbouring frequencies. If the window width is too large, then in time domain it may capture several heart beats (which may include an unhealthy heartbeat). Thus, the FT of this sample will also contain the effects of unhealthy heartbeat.

d)

- (i) The range of values $\rho(m)$ can take is $[-1, 1]$.
- (ii) Normalization ensures that the score measures only the shape similarity and not the amplitudes (closer to +1 means almost overlapping signals and -1 means that the signal is almost opposite to what it should have been i.e. similar shape but flipped about x axis). If the scores were unnormalized, then results can be misleading. For example, if a heartbeat was healthy but twice as tall as the sampled heartbeat, then the score will be doubled. It leads to logging of false unhealthy heartbeats.
- (iii) For arrhythmia inverted heartbeat, the shape of the heartbeat remains the same, but it gets flipped about the x axis. Hence, the expected value of the score will be close to -1 where - indicates flipped signal and 1 denotes the shape like healthy heartbeat. It is easier to flag by catching highly negative dips in the correlation plot.

e)

- (i) The onset is declared if the score deviates too much from its normal value of 1 i.e. $p(m) < 0.5$. If the threshold is kept too high (like $p(m) < 0.95$), it triggers false alarms by logging healthy heartbeats with slightly deviating score as the onset. If the threshold is too low (like $p(m) < 0.2$), we can miss the actual onset by classifying unhealthy heartbeats with slightly higher scores as healthy.
- (ii) A healthy heartbeat display highly stable, uniformly spaced, solid horizontal harmonic bands in frequency domain (because FT of a periodic signal is impulse train). In Arrhythmia region, the neat horizontal bands break up, showing chaotic, fractured spectral components, shifting frequencies, and vertical disruptions where the clean periodicity drops out.
- (iii) The correlation plot is computed for each sample by sliding the template sample by sample in time domain and calculating the score for each sample. The spectrogram traverses the time domain in chunks of samples and calculates the FT of each chunk. The onset can be determined with an accuracy of 1 sample in case of correlation plot while the accuracy of spectrogram will be much lower with the maximum error equal to number of samples in each chunk.

f)

- (i) By Nyquist theorem,
 $f_s \geq 2 \cdot f_{\max} = 2 \cdot 40 = 80 \text{ Hz}$
- (ii) If $f_s = 50 \text{ Hz}$, the $f_{\max} = 25 \text{ Hz}$. Since the QRS spikes correspond to $f = 40 \text{ Hz}$, the frequency in the region 25 to 40 Hz will undergo aliasing and as a result they get blurred. Since correlation plot rely on shape, it will give wrong results.
- (iii) Before down sampling, the signal passes through an anti-aliasing filter which makes $f_{\max} = 25 \text{ Hz}$ by making all the frequencies between 25 Hz and 40 Hz zero. The cost of this fix is the permanent loss of fine clinical diagnostic detail. By eliminating all frequencies between 25 Hz and 40 Hz, the highest-speed information in the heartbeat gets discarded.

g) The Arrhythmic heartbeat first occurs at beat index 12

h) $n_{\text{perseg}} = 600$. Taking a higher value results in higher error in time domain while taking a lower value results in a lower resolution in the frequency domain.

